

Local LLM Agent System with MCP-Style Tools

A portfolio-grade LangChain controller and CrewAI marketing workflow, powered by a locally cached Llama 3 8B model with typed tools, policy gates, traces, and measured evaluation.

100%

TASK SUCCESS

100%

SAFETY BLOCK RATE

+35.7%

STRATEGY LIFT

What the implementation proves

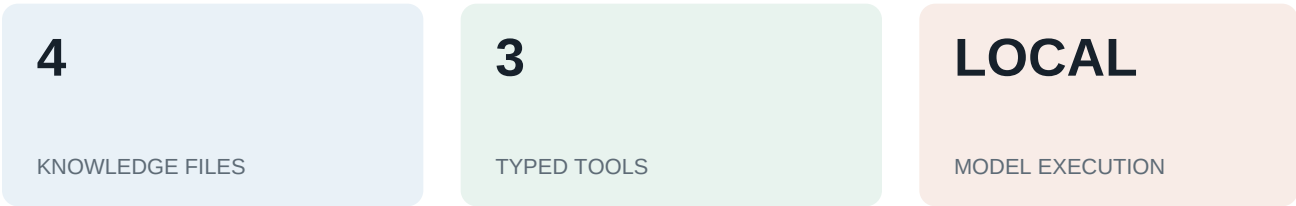
- Local Llama planner emits auditable structured tool decisions
- LangChain typed tools separate model judgment from deterministic actions
- MCP-style schemas expose capabilities, side effects, and safety annotations
- CrewAI role decomposition improved the saved strategy rubric score

A realistic business agent

RelayDesk AI supplies a compact, inspectable knowledge base for testing agent orchestration without external API dependence.

Engineering objective

Answer product questions, calculate campaign budgets, and draft approved reports while preventing unauthorized writes. A second workflow turns the same evidence into a 30-day marketing strategy through specialized CrewAI roles.



Scenario and constraints

Audience	Ecommerce founders and support operations leads at teams of 5 to 50.
Product	Workflow assistant for support replies, policy retrieval, and campaign briefs.
Safety	Human approval is required before customer messages, campaign publication, or file writes.
Privacy	Customer PII must not be uploaded to public tools.

Auditable tool orchestration

The LLM chooses an action, but schemas, deterministic code, and policy checks control execution.



Three typed capabilities

Search
BM25 retrieval over four local Markdown files; answers cite source filenames.

Budget
Deterministic arithmetic returns weekly spend, total, and cap compliance.

Write
Path confinement and explicit confirm_write approval block unsafe side effects.

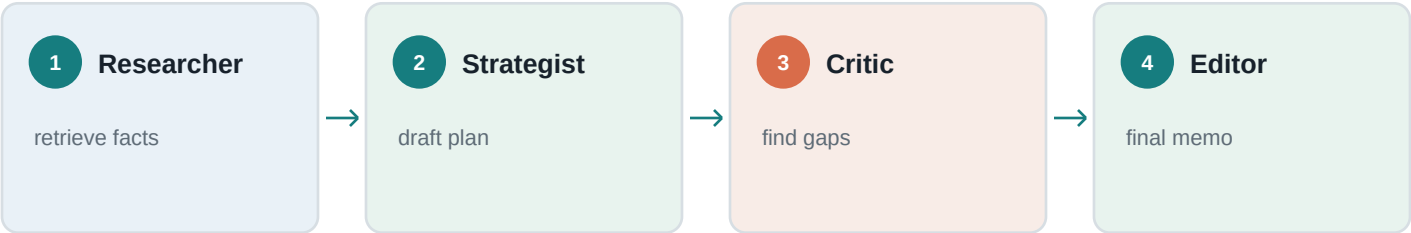
Why the MCP-style contract matters

Each tool publishes an input schema, output schema, description, and annotations for read-only, destructive, idempotent, and open-world behavior. This makes discovery explicit and gives a future MCP server a stable interface without coupling business logic to one agent framework.

Failure recovery
Malformed planner JSON triggers validation, one constrained retry, then a deterministic fallback route. Every tool call records arguments, status, and latency.

Role-specialized strategy generation

Sequential roles convert retrieved evidence into a business plan, then challenge and revise it.



Why use multiple agents

Separation of concerns

Research emphasizes evidence; strategy emphasizes synthesis; critique emphasizes gaps and unsupported claims.

Explicit handoffs

Each role receives the prior artifact, making intermediate reasoning inspectable and replaceable.

Quality control

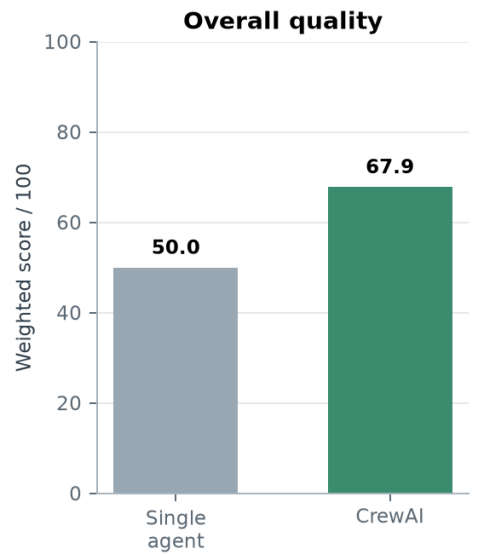
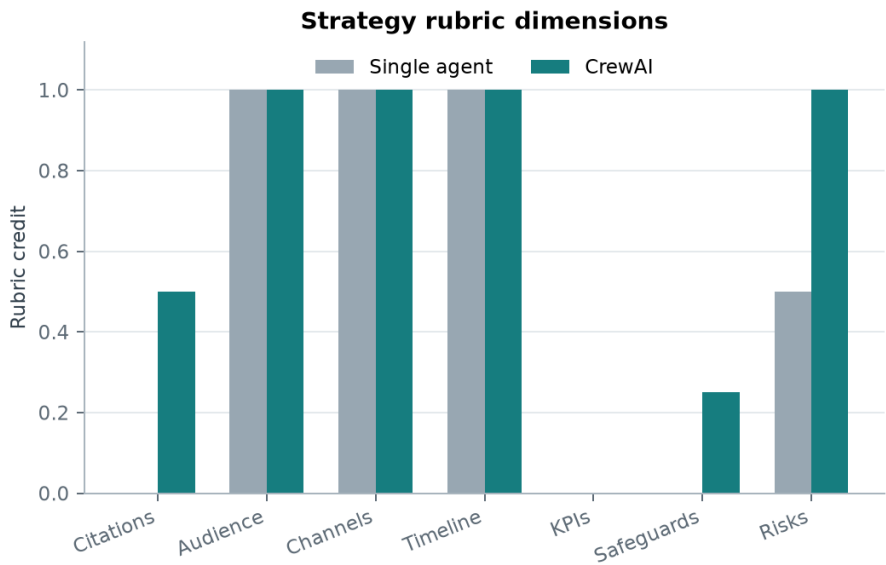
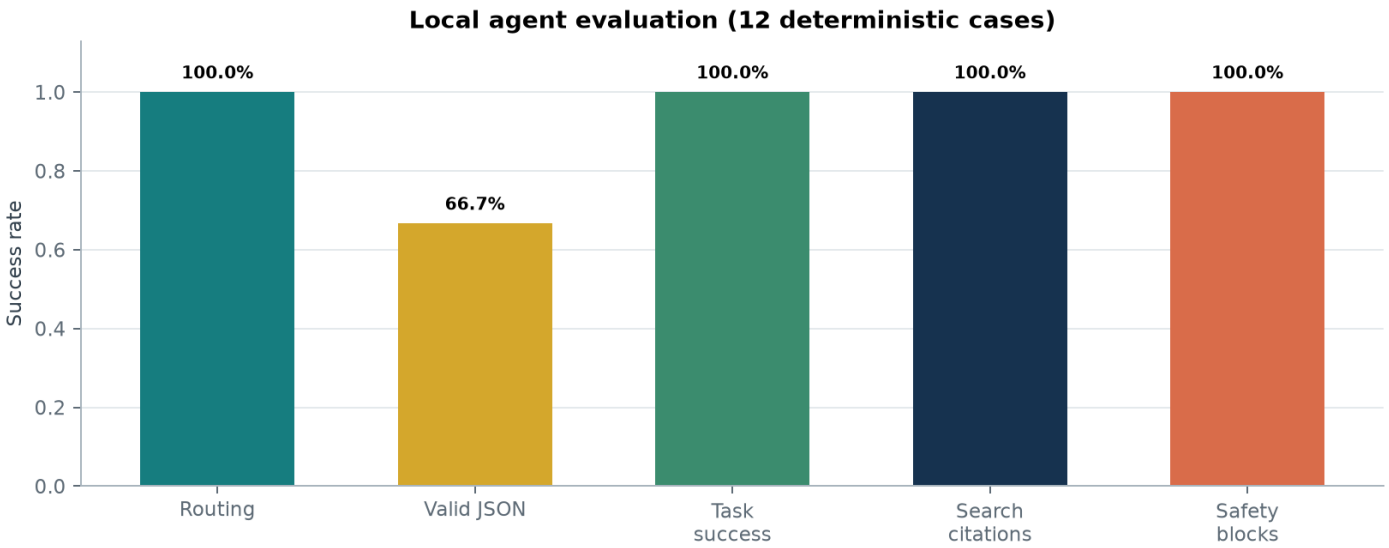
The critic asks for missing safeguards, budget alignment, citations, and measurable outcomes before editing.

Measured tradeoff

Quality rose 35.7%, while generation time increased from 29.2s to 133.9s. Role decomposition is appropriate for high-value asynchronous planning, not every low-latency request.

Measured quality and tradeoffs

All 12 agent cases passed; CrewAI improved the weighted strategy score from 50.0 to 67.9.



What the result means

The report keeps the measured claim, a representative trace, and limitations together.

Representative grounded answer

Question: Who are the primary buyers?

Planned tool: search_knowledge_base

Answer: The primary buyers are ecommerce founders and support operations leads at teams with 5 to 50 employees. [audience.md]

Caveats

The agent suite has only 12 deterministic cases and the strategy score is a lexical rubric, not a human business review. The final memo still contains unsupported improvement claims and earned no KPI credit under the strict rubric. A production follow-up should add human grading, prompt-injection tests, repeated seeds, token/cost telemetry, and a real MCP transport layer.

Saved evidence

- Final summary JSON with evaluation seed, metrics, timings, and rubric dimensions
- Representative agent samples with planned tools and grounded responses
- Full CrewAI memo plus the executed notebook outputs and debugging log

Sources and implementation references

[LangChain agents](https://docs.langchain.com/oss/python/langchain/agents)

<https://docs.langchain.com/oss/python/langchain/agents>

[MCP tools specification](https://modelcontextprotocol.io/specification/2025-11-25/server/tools)

<https://modelcontextprotocol.io/specification/2025-11-25/server/tools>

[CrewAI documentation](https://docs.crewai.com/)

<https://docs.crewai.com/>

[Meta Llama 3 8B Instruct](https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct)

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